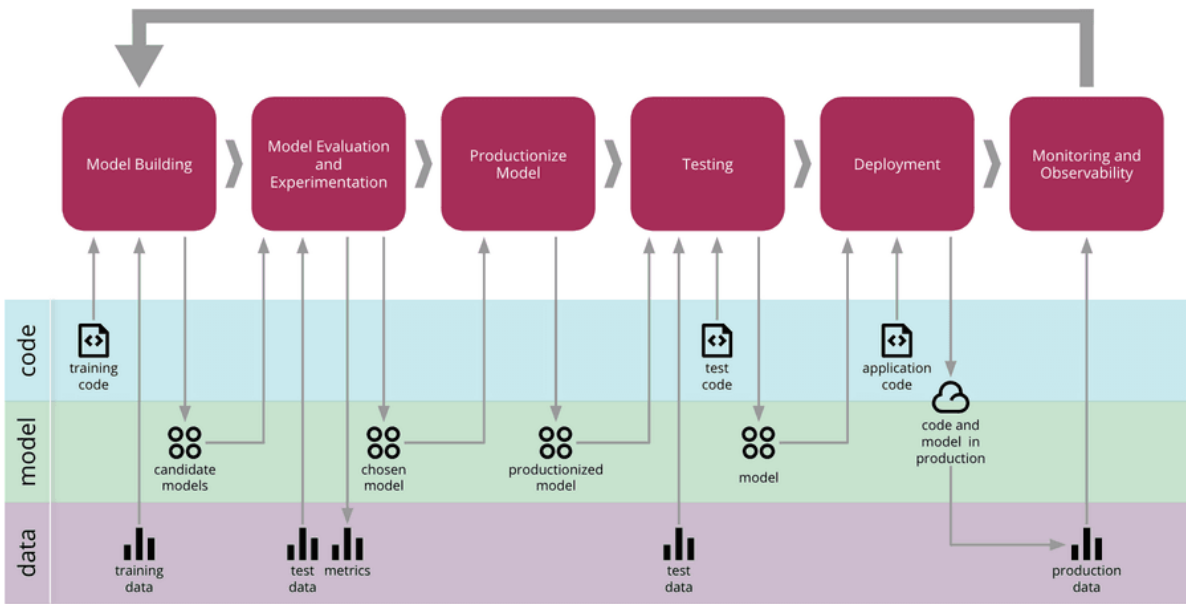


WHEN YOU'VE PREPARED JUST DSA FOR INTERVIEW



Agenda

- * What does ML system design includes.



TESTING → **OFFLINE** → MSE, R² Score, MAE, MAPE,
ONLINE → CTR, Conversion, Return on Ad Spend
FUNCTIONAL → Throughput, Latency, Cost, Server Load, Memory Usage.

GOIBIBO → AI Algorithms - 5% traffic
 Classical NON-ML - 95% traffic

Which of these is NOT a system health/performance metric typically monitored after deploying an ML model?

0 users have participated

A	Requests per second	0%
✓ B	Accuracy of the model's predictions	0%
C	CPU/memory/data utilization	0%
D	Latency per request	0%

End Quiz Now



What is the purpose of the 'Model Evaluation and Experimentation' phase in the ML model lifecycle?

0 users have participated

A	Deploying the model to a production environment	0%
B	Preparing data and initial code formulation	0%
✓ C	Feature selection, hyperparameter tuning, and algorithm comparison	0%
D	Monitoring and observing the model in a live environment	0%

End Quiz Now



Drift In Model/Data

→ Something has changed → Target
→ Features

① Concept drift: properties of target has changed.

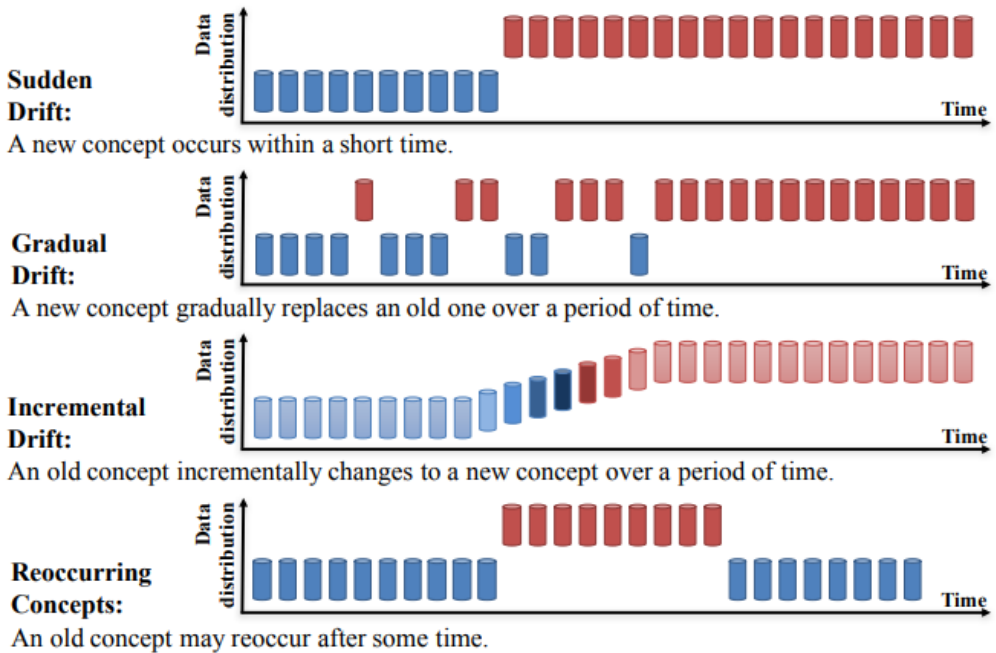
②. Fraud detection

③. Work hours → churn rate

Soln: Train model with more recent data

② Data Drift: Properties of input feature change

Soln: Create better Features.



Reasons for data drift

- changes in the data collection process
- changes in the data sources which provide the inputs
- changes in the business needs.
- Change in the distribution of the label/features in the data

How to Detect Drift!?

① Visual Inspection $\rightarrow$ MLFlow $\rightarrow$ metrics are changing over time.

monitor for distributions μ, σ

② Hypothesis testing $\rightarrow$ χ^2 test over past 6 months.

What is drift in the context of machine learning?

0 users have participated

- ☐ A The movement of an autonomous vehicle 0%
- ☐ B The physical displacement of a computing server 0%
- ☒ C The change in the statistical properties of model data over time 0%
- ☐ D The time it takes for a model to make a prediction 0%

[End Quiz Now](#)



1	K. Janani	239.56	164.57
2	Sanidhya Shukla	197.97	97.97
3	Yadwinder kaur	93.60	93.60
4	J. Ramesh Reddy	90.90	90.90
5	Rishikesh Kumar Jha	85.67	85.67
6	Ashirban	81.90	81.90
7	Kuntal Kule	70.17	70.17

Jan $\rightarrow$ trained, Prediction - Feb

└──────────────────┘

Drift

Metrics $\rightarrow$

Feb	1st	week
4	2nd	"
Feb	3rd	"

What type of drift occurs when there's a change in the relationship between input data 'x' and output data 'y'?

- 0 users have participated
- A Data drift 0%
 - B Business drift 0%
 - C Algorithm drift 0%
 - ☒ D Concept drift 0%

[End Quiz Now](#)

P 2			AS 1			J 3		
♂ Prashant Barkale 4/14 ⚡ 298.33			♂ ARIJIT SUR 4/14 ⚡ 361.36			♂ Jigar 3/14 ⚡ 230.56		
4	K. Janani	2/4	⚡	184.57				
5	J. Ramesh Reddy	2/4	⚡	179.66				
6	Rishikesh Kumar Jha	2/4	⚡	177.60				
7	Kuntal Kole	2/4	⚡	151.06				
8	Sanidhya Shukla	1/4	⚡	97.97				
9	Yadwinder kaur	1/4	⚡	93.60				
10	Anirban	1/4	⚡	81.90				